

Wireshark Lab: DNS v7.0

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ANSWERS:

1. IP Address: 10.59.121.144#53
2. Server: 10.59.121.244
Address: 10.59.121.244#53
Non-authoritative answer:
birmingham.ac.uk nameserver = ns0cc1.bham.ac.uk.
birmingham.ac.uk nameserver = ns0ab.bham.ac.uk.
Authoritative answers can be found from:
ns0ab.bham.ac.uk internet address = 194.80.24.26
3. IP Address:
Server: 8.8.8.8
Address: 8.8.8.8#53

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[devmbandhiya@Devs-MacBook-Air ~ % nslookup www.nus.edu.sg

Server:          10.59.121.144
Address:         10.59.121.144#53

Non-authoritative answer:
www.nus.edu.sg canonical name = mgnzsqc.x.incapdns.net.
Name:   mgnzsqc.x.incapdns.net
Address: 45.60.35.225

[devmbandhiya@Devs-MacBook-Air ~ % nslookup -type=NS birmingham.ac.uk
Server:          10.59.121.144
Address:         10.59.121.144#53

Non-authoritative answer:
birmingham.ac.uk      nameserver = ns0ab.bham.ac.uk.
birmingham.ac.uk      nameserver = ns0cc1.bham.ac.uk.

Authoritative answers can be found from:
ns0ab.bham.ac.uk      internet address = 194.80.24.26
ns0cc1.bham.ac.uk      internet address = 194.80.24.5

[devmbandhiya@Devs-MacBook-Air ~ % nslookup -type=MX yahoo.com 8.8.8.8

Server:          8.8.8.8
Address:         8.8.8.8#53

Non-authoritative answer:
yahoo.com      mail exchanger = 1 mta6.am0.yahoodns.net.
yahoo.com      mail exchanger = 1 mta7.am0.yahoodns.net.
yahoo.com      mail exchanger = 1 mta5.am0.yahoodns.net.
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4. The DNS query and response messages are sent over UDP.
5. Both the destination port of DNS query message and source port of DNS response message is same i.e. 53.
6. IP Address: 10.59.121.144
Yes, the two IP Addresses are the same.
7. DNS Query Type: A (which means address)
Answers: 0
8. Response message contains 2 answers. It contains the following information:
www.ietf.org: type A, class IN, addr 104.16.45.99
www.ietf.org: type A, class IN, addr 104.16.44.99
9. Dst: 104.16.45.99 -> It corresponds to the IP in the DNS response.
10. No, the host does not issue new queries. The images have the same domain as the webpage.

ip.addr == 10.70.48.108						
No.	Time	Source	Destination	Protocol	Length	Info
294	34.597236	10.70.48.108	104.16.45.99	UDP	639	49626 → 443 Len=597
295	34.907444	10.70.48.108	104.16.45.99	UDP	639	49626 → 443 Len=597
296	35.532037	10.70.48.108	104.16.45.99	UDP	639	49626 → 443 Len=597
297	36.776594	10.70.48.108	104.16.45.99	UDP	639	49626 → 443 Len=597
298	38.833647	10.70.48.108	104.16.45.99	UDP	83	49626 → 443 Len=41
299	39.837904	10.70.48.108	104.16.45.99	UDP	83	49626 → 443 Len=41
300	40.842401	10.70.48.108	104.16.45.99	UDP	108	49626 → 443 Len=66
301	40.848349	10.70.48.108	10.59.121.144	DNS	72	Standard query 0x7dcb HTTPS www.ietf.org
302	40.848507	10.70.48.108	10.59.121.144	DNS	72	Standard query 0x1870 A www.ietf.org
303	40.854414	10.59.121.144	10.70.48.108	DNS	117	Standard query response 0x7dcb HTTPS www.ietf.org HTTPS
304	40.854416	10.59.121.144	10.70.48.108	DNS	104	Standard query response 0x1870 A www.ietf.org A 104.16.45.99 A 104.16.44.99
305	40.856249	10.70.48.108	104.16.45.99	TCP	78	52021 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=129806071
306	40.864779	104.16.45.99	10.70.48.108	TCP	66	443 → 52021 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1382 SACK_PERM
307	40.865346	10.70.48.108	104.16.45.99	TCP	54	52021 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
308	40.865350	10.70.48.108	104.16.45.99	TLSv1...	571	Client Hello (SNI=www.ietf.org)
309	40.871073	104.16.45.99	10.70.48.108	TCP	60	443 → 52021 [ACK] Seq=1 Ack=518 Win=73728 Len=0
310	40.871898	104.16.45.99	10.70.48.108	TLSv1...	1436	Server Hello, Change Cipher Spec
311	40.871900	104.16.45.99	10.70.48.108	TCP	1436	443 → 52021 [ACK] Seq=1383 Ack=518 Win=73728 Len=1382 [TCP PDU reassem
312	40.871902	104.16.45.99	10.70.48.108	TLSv1...	208	Application Data
313	40.873139	10.70.48.108	104.16.45.99	TCP	54	52021 → 443 [ACK] Seq=518 Ack=2919 Win=262144 Len=0
314	40.881642	10.70.48.108	104.16.45.99	TLSv1...	118	Change Cipher Spec, Application Data

> Frame 304: 104 bytes on wire (832 bits), 104 bytes captured (832 bits) on interface en0, id 0	0000	1c 57 dc 4d fa 19 ec 67	94 fb 01 00 08 00
> Ethernet II, Src: HewlettPacka_fb:01:00 (ec:67:94:fb:01:00), Dst: Apple_4d:fa:19 (1c:57:dc:4d:fa:19)	0010	00 5a 3b d6 00 00 7f 11	41 40 0a 3b 79 99
> Internet Protocol Version 4, Src: 10.59.121.144, Dst: 10.70.48.108	0020	30 6c 00 35 cc 25 00 46	0d 52 18 70 81 80
> User Datagram Protocol, Src Port: 53, Dst Port: 52261	0030	00 02 00 00 00 00 03 77	77 77 04 69 65 7f
> Domain Name System (response)	0040	6f 72 67 00 00 01 00 01	c0 0c 00 01 00 00
Transaction ID: 0x1870	0050	00 fa 00 04 68 10 2d 63	c0 0c 00 01 00 00
> Flags: 0x180 Standard query response, No error	0060	00 fa 00 04 68 10 2c 63	
Questions: 1			
Answer RRs: 2			
Authority RRs: 0			
Additional RRs: 0			
> Queries			
> Answers			
> www.ietf.org: type A, class IN, addr 104.16.45.99			
> www.ietf.org: type A, class IN, addr 104.16.44.99			
[Request In: 302]			
[Time: 0.005909000 seconds]			

11. Both the destination port of DNS query message and source port of DNS response message is same i.e. 53.

12. IP Address: 10.59.121.144 Yes, the two IP Addresses are the same.

13. DNS Query Type: A (which means address) and CNAME

Answers: 0

14. Response message contains 2 answers. It contains the following information:

www.mit.edu: type CNAME, class IN, cname

www.mit.edu.edgekey.net

www.mit.edu.edgekey.net: type CNAME, class IN, cname

e9566.dscb.akamaiedge.net

e9566.dscb.akamaiedge.net: type A, class IN, addr 23.58.1.151

dns

Time	Source	Destination	Protocol	Length	Info
31	11.595509	10.70.48.108	10.59.121.144	DNS	80 Standard query 0x118c HTTPS clients.l.google.com
32	11.599543	10.59.121.144	10.70.48.108	DNS	130 Standard query response 0x118c HTTPS clients.l.google.com SOA ns1.google.com
39	14.171192	10.70.48.108	10.59.121.144	DNS	80 Standard query 0xe23e HTTPS cdn2.smoot.apple.com
40	14.171408	10.70.48.108	10.59.121.144	DNS	80 Standard query 0x0bbd A cdn2.smoot.apple.com
42	14.176220	10.59.121.144	10.70.48.108	DNS	168 Standard query response 0xe23e HTTPS cdn2.smoot.apple.com CNAME cdn.smoot.g.aaplimg.
43	14.176221	10.59.121.144	10.70.48.108	DNS	146 Standard query response 0x0bbd A cdn2.smoot.apple.com CNAME cdn.smoot.g.aaplimg.com
44	14.176945	10.70.48.108	10.59.121.144	DNS	83 Standard query 0x6c77 HTTPS cdn.smoot.g.aaplimg.com
45	14.181035	10.59.121.144	10.70.48.108	DNS	143 Standard query response 0x6c77 HTTPS cdn.smoot.g.aaplimg.com SOA a.gslb.aaplimg.com
05	15.572571	10.70.48.108	10.59.121.144	DNS	71 Standard query 0x7524 HTTPS www.mit.edu
06	15.572798	10.70.48.108	10.59.121.144	DNS	71 Standard query 0xe977 A www.mit.edu
07	15.703863	10.59.121.144	10.70.48.108	DNS	160 Standard query response 0xe977 A www.mit.edu CNAME www.mit.edu.edgekey.net CNAME e95
08	15.733907	10.59.121.144	10.70.48.108	DNS	144 Standard query response 0x7524 HTTPS www.mit.edu CNAME www.mit.edu.edgekey.net CNAME
09	15.734558	10.70.48.108	10.59.121.144	DNS	85 Standard query 0xb6e6 HTTPS e9566.dscb.akamaiedge.net
10	15.740362	10.59.121.144	10.70.48.108	DNS	149 Standard query response 0xb6e6 HTTPS e9566.dscb.akamaiedge.net SOA n0dscb.akamaiedge
43	16.362468	10.70.48.108	10.59.121.144	DNS	84 Standard query 0x9f69 HTTPS www.googletagmanager.com
44	16.362663	10.70.48.108	10.59.121.144	DNS	84 Standard query 0xe80f A www.googletagmanager.com
48	16.367455	10.59.121.144	10.70.48.108	DNS	100 Standard query response 0xe80f A www.googletagmanager.com A 142.250.195.72
49	16.367456	10.59.121.144	10.70.48.108	DNS	141 Standard query response 0x9f69 HTTPS www.googletagmanager.com SOA ns1.google.com
18	16.741361	10.70.48.108	10.59.121.144	DNS	75 Standard query 0xd399 HTTPS use.typekit.net
19	16.741509	10.70.48.108	10.59.121.144	DNS	75 Standard query 0xc0cb A use.typekit.net
83	16.747749	10.59.121.144	10.70.48.108	DNS	216 Standard query response 0xd399 HTTPS use.typekit.net CNAME use-styles.adobe.com.edges

> Frame 407: 160 bytes on wire (1280 bits), 160 bytes captured (1280 bits) on interface en0, id 0

> Ethernet II, Src: HewlettPacka_fb:01:00 (ec:67:94:fb:01:00), Dst: Apple_4d:fa:19 (1c:57:dc:4d:fa:19)

> Internet Protocol Version 4, Src: 10.59.121.144, Dst: 10.70.48.108

> User Datagram Protocol, Src Port: 53, Dst Port: 55256

▼ Domain Name System (response)

Transaction ID: 0xe977

> Flags: 0x8180 Standard query response, No error

Questions: 1

Answer RRs: 3

Authority RRs: 0

Additional RRs: 0

> Queries

▼ Answers

> www.mit.edu: type CNAME, class IN, cname www.mit.edu.edgekey.net

> www.mit.edu.edgekey.net: type CNAME, class IN, cname e9566.dscb.akamaiedge.net

> e9566.dscb.akamaiedge.net: type A, class IN, addr 23.58.1.151

[Request In: 406]

[Time: 0.131065000 seconds]

0000 1c 57 dc 4d fa 19 ec 67 94 fb 01 00 08 00 45 00

0010 00 92 3c 83 00 00 7f 11 40 5b 0a 3b 79 90 0a 46

0020 30 6c 00 35 d7 d8 00 7e c1 63 e9 77 81 80 00 01

0030 00 03 00 00 00 00 03 77 77 77 03 6d 69 74 03 65

0040 64 75 00 00 01 00 01 c0 0c 00 05 00 01 00 00 03

0050 e7 00 19 03 77 77 77 03 6d 69 74 03 65 64 75 07

0060 65 64 67 65 6b 65 79 03 6e 65 74 00 c0 29 00 05

0070 00 01 00 00 00 3c 00 18 05 65 39 35 36 36 04 64

0080 73 63 62 0a 61 6b 61 6d 61 69 65 64 67 65 c0 3d

0090 c0 4e 00 01 00 01 00 00 00 14 00 04 17 3a 01 97

16. The DNS query sent the message to IP Address: 10.59.121.144
This IP Address is the default local DNS server.
17. DNS Query Type: CNAME and A
Answers: 0
18. The DNS response message provides 8 nameservers. They are as follows:
- mit.edu nameserver = ns1-37.akam.net
 - mit.edu nameserver = ns1-173.akam.net
 - mit.edu nameserver = eur5.akam.net
 - mit.edu nameserver = use5.akam.net
 - mit.edu nameserver = usw2.akam.net
 - mit.edu nameserver = asia1.akam.net
 - mit.edu nameserver = asia2.akam.net
 - mit.edu nameserver = use2.akam.net

use5.akam.net internet address = 2.16.40.64

Only for one of the servers, the IP address is provided.

The image shows a Wireshark packet capture of a DNS response. The packet list on the left shows a DNS response from 10.59.121.144 to 10.70.42.73. The packet details pane shows the response structure, including the transaction ID, flags, and a list of 8 nameservers for mit.edu. The packet bytes pane shows the raw data of the response.

Transaction ID: 0x0004

Flags: 0x8180 Standard query response, No error

Questions: 1

Answer RRs: 8

Authority RRs: 0

Additional RRs: 1

Queries

Answers

- mit.edu: type NS, class IN, ns ns1-37.akam.net
- mit.edu: type NS, class IN, ns ns1-173.akam.net
- mit.edu: type NS, class IN, ns eur5.akam.net
- mit.edu: type NS, class IN, ns use5.akam.net
- mit.edu: type NS, class IN, ns usw2.akam.net
- mit.edu: type NS, class IN, ns asia1.akam.net
- mit.edu: type NS, class IN, ns asia2.akam.net
- mit.edu: type NS, class IN, ns use2.akam.net

Additional records

[Request In: 603]

[Time: 0.003826000 seconds]

Domain Name System: Protocol

Packets: 658 · Displayed: 40 (6.1%) · Dropped: 0 (0.0%)

Profile: Default

20. The DNS query sent the message to IP Address: 10.59.121.144
This IP Address is the default local DNS server.
21. DNS Query Type: CNAME and A
Answers: 0
22. 1 answer is provided. It contains the following:
www.aiit.or.kr: type A, class IN, addr 58.229.6.225

ip.addr == 23.58.1.151						
No.	Time	Source	Destination	Protocol	Length	Info
10	3.592364	10.70.48.108	10.59.121.144	DNS	74	Standard query 0x18fb HTTPS www.aiit.or.kr
11	3.592652	10.70.48.108	10.59.121.144	DNS	74	Standard query 0x3267 A www.aiit.or.kr
12	3.600378	10.59.121.144	10.70.48.108	DNS	90	Standard query response 0x3267 A www.aiit.or.kr A 58.229.6.225
38	4.076491	10.70.48.108	10.59.121.144	DNS	79	Standard query 0xe930 HTTPS ajax.googleapis.com
39	4.076630	10.70.48.108	10.59.121.144	DNS	79	Standard query 0x98da A ajax.googleapis.com
40	4.076902	10.70.48.108	10.59.121.144	DNS	77	Standard query 0x3022 HTTPS spi.maps.daum.net
41	4.076968	10.70.48.108	10.59.121.144	DNS	77	Standard query 0xe6ca A spi.maps.daum.net
42	4.080753	10.59.121.144	10.70.48.108	DNS	136	Standard query response 0xe930 HTTPS ajax.googleapis.com SOA ns1.google.com
43	4.080754	10.59.121.144	10.70.48.108	DNS	95	Standard query response 0x98da A ajax.googleapis.com A 142.250.196.170
96	4.223620	10.59.121.144	10.70.48.108	DNS	128	Standard query response 0x18fb HTTPS www.aiit.or.kr SOA ns9.dnszi.com
120	4.344765	10.59.121.144	10.70.48.108	DNS	134	Standard query response 0xe6ca A spi.maps.daum.net CNAME spi-maps-5d3b60w9.kg
172	4.505970	10.59.121.144	10.70.48.108	DNS	118	Standard query response 0x3022 HTTPS spi.maps.daum.net CNAME spi-maps-5d3b60w9.kg
173	4.506925	10.70.48.108	10.59.121.144	DNS	87	Standard query 0x5ae3 HTTPS spi-maps-5d3b60w9.kgslb.com
174	4.511717	10.59.121.144	10.70.48.108	DNS	151	Standard query response 0x5ae3 HTTPS spi-maps-5d3b60w9.kgslb.com SOA gsx-gs
325	5.100267	10.70.48.108	10.59.121.144	DNS	74	Standard query 0x05ab HTTPS t1.daumcdn.net
328	5.100456	10.70.48.108	10.59.121.144	DNS	74	Standard query 0x7069 A t1.daumcdn.net
336	5.158317	10.59.121.144	10.70.48.108	DNS	214	Standard query response 0x7069 A t1.daumcdn.net CNAME t1-wq2vgaia.kgslb.com

> Frame 12: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface en0,	0000	1c 57 dc 4d fa 19 ec 67 94 fb 01 00 08 00 45 00	·W·M··g
> Ethernet II, Src: HewlettPacka_fb:01:00 (ec:67:94:fb:01:00), Dst: Apple_4d:fa:19 (1c:57:dc:4d:fa:19)	0010	00 4c 42 b9 00 00 7f 11 3a 6b 0a 3b 79 90 0a 46	·LB·...·
> Internet Protocol Version 4, Src: 10.59.121.144, Dst: 10.70.48.108	0020	30 6c 00 35 d7 be 00 38 68 5d 32 67 81 80 00 01	01·5·...8
> User Datagram Protocol, Src Port: 53, Dst Port: 55230	0030	00 01 00 00 00 00 03 77 77 77 04 61 69 69 74 02	w
> Domain Name System (response)	0040	6f 72 02 6b 72 00 00 01 00 01 c0 0c 00 01 00 01	or·kr·...
Transaction ID: 0x3267	0050	00 00 0a 52 00 04 3a e5 06 e1	...R·:..
> Flags: 0x8180 Standard query response, No error			
Questions: 1			
Answer RRs: 1			
Authority RRs: 0			
Additional RRs: 0			
> Queries			
> Answers			
> www.aiit.or.kr: type A, class IN, addr 58.229.6.225			
[Request In: 11]			
[Time: 0.007726000 seconds]			